

native_triton_sparse_attn_decode(*q*, *blocks*, *routing*)

HAS_TRITON ?

no

yes

@triton.autotune over (S_MAX, BLOCKS_PER_CHUNK, num_warps)
first five real-shape calls benchmarked → fastest config locked in

_fused_sparse_decode_kernel — one launch grid

Triton GPU Kernel Execution Steps

1. Anchor Score: $s_{\text{anc}} = (q \cdot a_k) \cdot \text{scale}$

2. Delta Projection: $\tilde{q} = qV_K^T \in \mathbb{R}^{32}$

3. Token Scores: $\delta s_i = \tilde{q} \cdot U_i$ (all i in block)

4. Scatter $q \cdot R_K$ into score at res_K_positions

5. Scatter $p \cdot R_V$ into numerator at res_V_positions

6. Online Softmax → accumulate O , lse

$O(K \cdot r \cdot B)$ compute — block is never expanded to dense $\hat{K}$

Fallback Path

_pytorch_vectorized_sparse_attn_decode()

DKV_TRITON_STRICT = 1 ?

yes

no

**hard error
(re-raise)**

PyTorch output
+ telemetry log